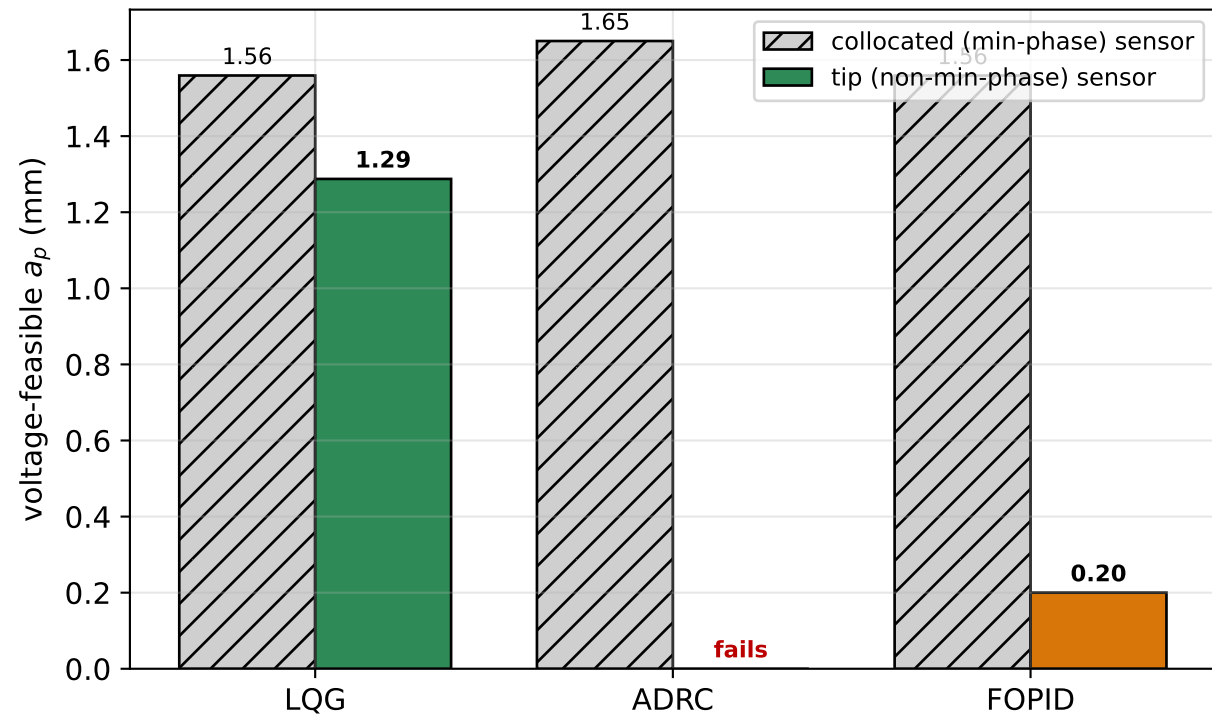


All controllers forced onto the tip (non-minimum-phase) sensor

(a) feasible depth on the common tip sensor
ADRC fails on the NMP tip; FOPID degrades; LQG least affected



(b) tip response at $a_p = 0.5$ mm on the tip sensor
diverging traces terminate at the chatter threshold

